

Battery components Analyses using AFM-in-SEM LiteScope™

An Overview of NenoVision's LiteScope Battery Materials Solutions

Performance and longevity of batteries is intrinsically linked to the materials used in their construction. This application note presents an overview of key material components within cell and battery technologies, such as cathodes, anodes, and electrolytes, in both solid and liquid states.

Enhanced by NenoVision's LiteScope, this article is intended to provide a foundational understanding of the raw active materials involved in battery technology, emphasising their roles in energy storage and transfer mechanisms.

Keywords or phrases: Battery, cathode, anode, electrolyte, AFM-in-SEM, NenoVision LiteScopeSWILE

Introduction

Energy storage devices are crucial to today's technology. Understanding their features, performance, aging, and capacity is vital.

Most high-end batteries are Li-ion, with active materials extremely susceptible to atmospheric contamination and degradation.

NenoVision's battery materials solution with LiteScope provides a complete hardware and measurement workflow approach. It allows for an air-free sample transfer system that prevents contamination by oxygen and humidity, while ensuring uncompromised sample surface preparation.

The imaging power of AFM-in-SEM provides a complex and site-specific understanding of the electrical and chemical properties of battery components.

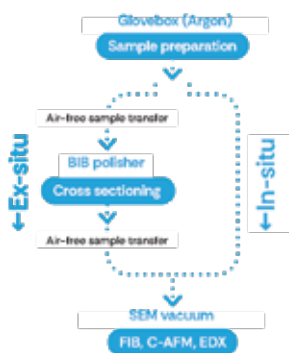
Measurement Workflow

1. Ex-situ sample preparation using Sample Transfer System

This option allows larger cross-section preparation in specialized instruments while maintaining the sample surface in the necessary environment, even during transfer and AFM-in-SEM measurement.

2. In-situ sample preparation using Focused Ion Beam

LiteScope enables safe AFM probe positioning for large stage tilts, preserving the tip from redeposition and allowing the use of FIB to cut a trench anywhere on the sample. This allows measurement of the uncovered surface using AFM-in-SEM technology.



Cathode Tape Inspection

Solid State Batteries (SSBs) show promise over Li-ion batteries with **higher energy density, longer lifespan, and improved safety**. A cathode tape composed of Lithium Nickel Manganese Cobalt Oxide (NMC) particles was opened after 200 cycles in a glovebox, cross-sectioned ex-situ, and measured using **AFM-in-SEM** (Measurement Workflow 1).

SEM imaging identified the polished area of interest. The EDS map showed the cathode tape's composition, including the solid electrolyte, NMC particles, and carbon additive. C-AFM confirmed the carbon additive's conductivity and its role in connectivity.

material functions or failures without altering the sample surface.

Key benefits and features

Fast and easy sample transfer: Small evacuation time is needed to exchange the samples.

No air or humidity exposure: Air-sensitive samples are protected from surface contamination or oxidation in a vacuum or inert gas atmosphere at all times.

Complete measurement workflow: A comprehensive procedure for the cross-sectional analysis of complex battery components and materials testing.

Upgradation to SEM solution: Extended possibilities are enabled using FIB/GIS prep, AFM, C-AFM, KPFM, phase imaging, SE, BSE, EDX, Fz spectroscopy.

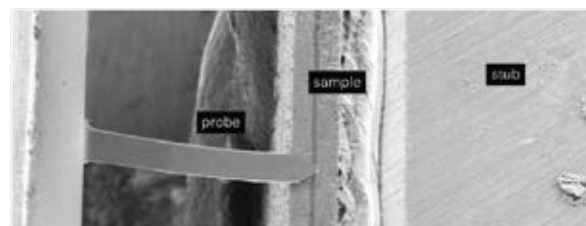
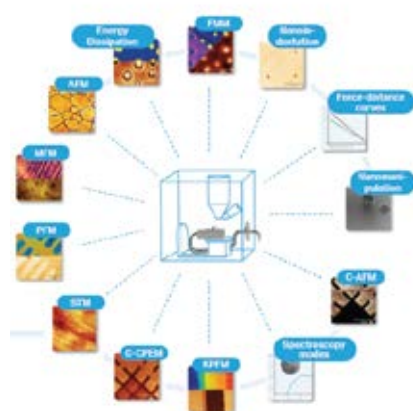


Figure 1: SEM overview: probe navigation to cross-sections



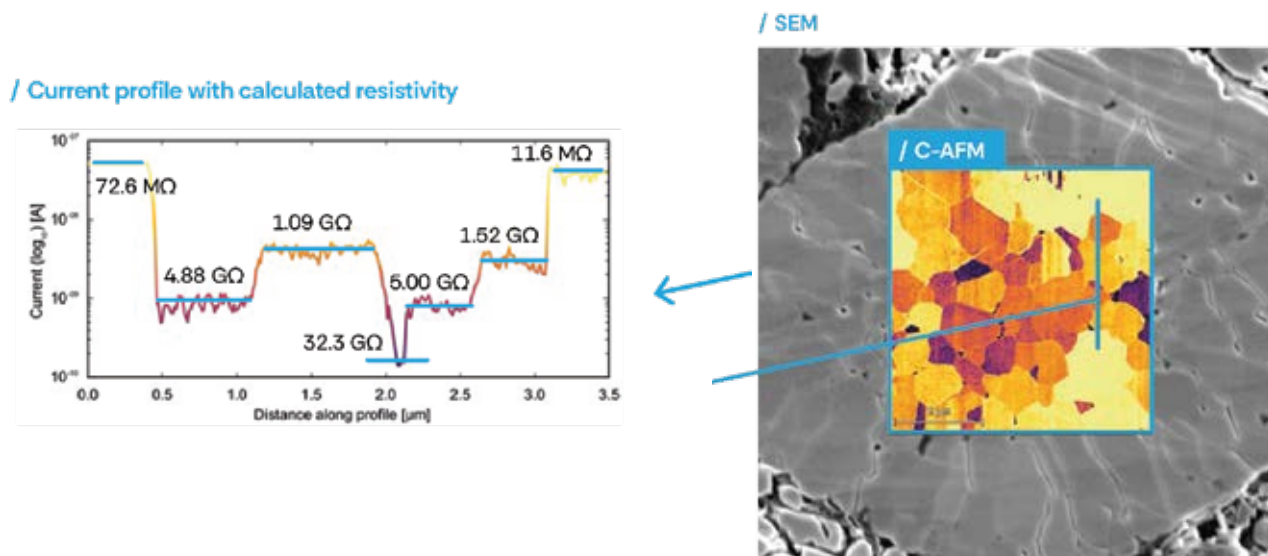


Figure 2: Complex conductivity distribution inside the secondary NMC particle is revealed by C-AFM

IntraParticle Conductivity

The cycling of batteries involves the transport of lithium ions from the cathode to the anode and vice versa, which causes the electrode particles to expand and contract. Over many cycles, the battery performance decreases. Understanding these degradation mechanisms is crucial for longer battery lifetime and better performance.

The C-AFM was employed to image a secondary particle from an NMC cathode of a liquid electrolyte (LE) battery after 200 charge-discharge cycles.

It revealed an interesting conductivity structure of primary particles, and the cracks formed between them during cycling. While there is an overarching tendency for the primary particles to be less conductive closer to the particle centre, another effect causes seemingly random conductivity variations.

Local C-AFM proves, that some of the primary particles lost electrical contact with their neighbours, probably due to cracks, and became disconnected. The effect is not negligible, as the current profile taken across a few grains shows differences in orders of magnitude, which then influences (decreases) overall battery performance.

Conclusion

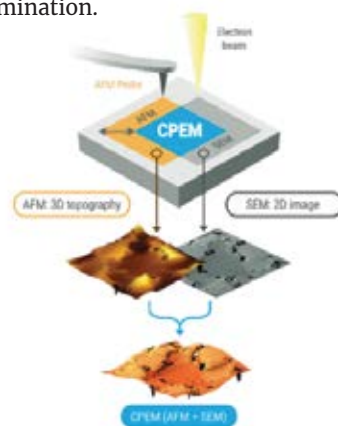
LiteScope™ solution for battery components analyses

Complete Workflow

Delivers a complete workflow for sample surface preparation, air-free sample transfer, site-specific electrical and chemical properties measurements, and data post-processing.

Battery components and materials testing

This setup enables the characterization of performance and failures, understanding of cycling effects, active material degradation, inter/ intraparticle connectivity, and delamination.



Application Showcase

Sample Transfer Module

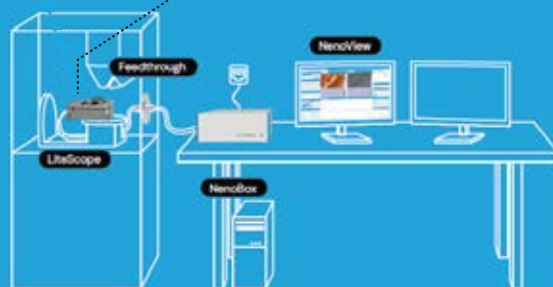
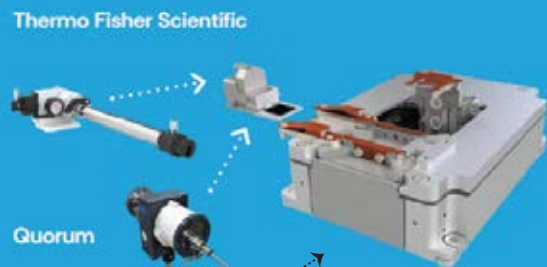
Dedicated Module for AFM-in-SEM LiteScope provides an effective solution for safe sample transfer under a controlled environment.

The module of compact design is compatible with most already existing sample transfer systems and their accessories.

LiteScope Setup

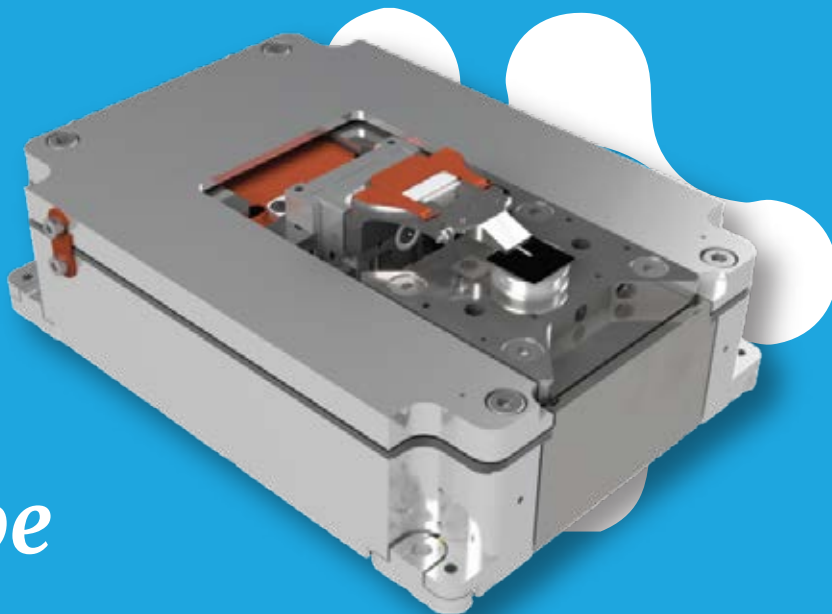
NenoVision combines tradition and expertise with unique solutions in nanoscale AFM-in-SEM microscopy.

Thanks to its optimized design, the AFM LiteScope seamlessly integrates into most SEM systems produced by Thermo Fisher Scientific, TESCAN, ZEISS, Hitachi, Jeol, and their accessories.



Application areas

- › Cell and battery components
- › Raw active materials
- › Cathodes
- › Anodes
- › Solid electrolytes
- › Liquid electrolytes



LiteScope

AFM-in-SEM

